# GORMLEY'S FORMULA SHEET

- **Present Value of Annuity**

$$PVA_0 = \frac{CF_1}{r} \times \left[1 - \frac{1}{(1+r)^n}\right]$$

- **Present Value of Growing Annuity**

$$PVA_0 = \frac{CF_1}{r-g} \times \left[1 - \left(\frac{1+g}{1+r}\right)^n\right]$$

- **Present Value of Perpetuity**

$$PVP_0 = \frac{CF_1}{r}$$

**PV of Growing Perpetuity**

$$PVP_0 = \frac{CF_1}{r-g}$$

- **Capital Asset Pricing Model (CAPM)**

$$r_i = r_f + \beta_i \left(r_m - r_f\right)$$

**Fundamental PE Ratio**

$$PE_F = \frac{b\left(1+g\right)}{r-g}$$

- **Free Cash Flows**

*FCF* = Unlevered net income + depreciation *(also called OCF)*
– Capital expenditures (*CapEx*)
– Change in net working capital (*ΔNWC*)

where

*OCF* = EBIT*(1 – tax rate) + depreciation
*CapEx* (definition #1) = End gross fixed assets – start gross fixed assets
*CapEx* (definition #2) = End NFA – start NFA + depreciation
*NWC* = current assets – current liabilities

- **Weighted Average Cost of Capital**

$$WACC = \left(\frac{E}{D+E}\right) r_e + \left(\frac{D}{D+E}\right) r_d \left(1 - T_C\right)$$

- **Firm Value using Discounted Cash Flows**

$$Firm\ value = cash_0 + \sum_{t=1}^{N}\left[\frac{FCF_t}{(1+WACC)^t}\right] + \frac{TV_N}{(1+WACC)^N}$$

- **Equity Value**

*Equity value = Firm value – debt – preferred equity*

- **M&A formula**

$PV_{AB} = PV_A + PV_B + Synergies$

## EXPLANATION OF ABOVE NOTATION

$CF_1$ = cash flow in year 1

$n$ = number of periods you will receive payments (e.g., in an annuity)

$r$ = discount rate

$r_i$ = discount rate on some generic asset $i$
$r_e$ = discount rate for equity (i.e., cost of equity)
$r_d$ = discount rate for debt (i.e., cost of debt)
$r_f$ = discount rate on a risk-free asset (i.e., return on risk-free asset)

$g$ = expected growth rate of the cash flows (e.g., in a perpetuity)

$b$ = what fraction of earnings per share (EPS) are paid out as dividends

$WACC$ = weighted average cost of capital

$FCF$ = free cash flow

$cash_0$ = excess cash-on-hand in year zero

$t$ = year t